

Osaid Khan

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EDUCATION

Pace University <i>Master of Science (MS) in Computer Science (GPA - 4.0/4.0)</i>	New York City, NY <i>Jan 2024 - Dec 2025</i>
Dr. A.P.J. Abdul Kalam Technical University <i>Bachelor of Technology (B.Tech) in Computer Science</i>	Noida, India <i>Jul 2016 - Sep 2020</i>

TECHNICAL SKILLS

Languages: Java, Python, Rust, JavaScript/TypeScript
AWS Services: EC2, RDS, S3, Lambda, CloudWatch, Step Functions, CloudFormation, CloudFront, API Gateway
Frameworks: PyTorch, FastAPI, FastMCP, LangGraph, LangChain, LangSmith, React, Sprint Boot
Databases: PostgreSQL, MongoDB, Redis
AI: Claude code, Codex, Pi coding-agent
Others: Google Cloud, Git, CI/CD, Kafka, Node.js

PROFESSIONAL EXPERIENCE

Software Engineer Sperse	Phoenix, AZ <i>Feb 2026 - present</i>
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- Designed and developed a **large-scale, multi-tiered** MCP AI interaction layer that translates natural-language intents into **safe, validated tool calls** across **1000+ platform endpoints**
- Implemented **complex software solutions** for developers (API-aware templates and code-generation workflows) to accelerate API integrations and deliver **efficient, reusable, and reliable code**
- Identified and resolved **performance bottlenecks** by optimizing prompt routing and agent concurrency, maximizing **KV-cache hit rate**, reducing latency by **60%**, and lowering API costs by **50%**
- Architected a **multi-threaded, concurrent multi-agent** orchestration system using **LangChain** and **LangGraph** for scalable tool execution and reliable coordination across workflows

AI Researcher Pace Artificial Intelligence Lab	New York City, NY <i>Sep 2025 - Dec 2025</i>
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- Implemented **training/inference lifecycles** for **StyleGAN** and **Stable Diffusion** utilizing **CUDA** and **PyTorch** on an **HPC cluster**, applying **optimization techniques** for **Machine Learning** workloads
- Utilized **vLLM** for model inference with **Hugging Face** deployment, leveraging knowledge of **LLM fundamentals** and **transformer architecture** to ensure secure and efficient model serving
- Mentored junior engineers and **+50** students on GPU programming with **CUDA** and **Mojo**, and on building **AI Agents** to contribute to building a strong engineering culture

Software Engineer Qualitest	Noida, India <i>Feb 2021 - Dec 2023</i>
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- Designed and developed a **distributed systems** Storage Service using **Python**, **PostgreSQL**, and **Redis** handling large-scale file operations in **S3** and **MinIO** via expiring signed URLs
- Collaborated with **cross-functional teams** to implement an Identity & Access Management platform using **React**, **Spring Boot**, and **AWS**, supporting thousands of users across 30+ applications
- Identified opportunities for process enhancement by automating long-running data import jobs using **AWS Step Functions**, preventing **AWS Lambda** timeouts and debugging production issues

PROJECTS

Dark Factory - Orchestrator for AI coding CLIs <i>Crossterm, Ratatui, Y-crdt</i>	github.com/ozzyozbourne/dark-factory
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- Enables full ownership of CLI IO via **PTY**, developing **efficient, reusable code** to check and log every command before execution by bash shell, ensuring secure **complex software systems**
- Allows secure **distributed** agentic loops across machines using **Y-crdt** to signal the orchestrator, broadcasting **real-time** updates to eliminate polling overhead